.sova-trace: Claim-Conditioned Evidence Fidelity for AI-System Behavior Records

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Abstract

AI-system traces are frequently treated as interchangeable logs even when they capture different event classes, causal structure, redactions, or trust assumptions. We introduce .sova-trace and a claim-conditioned fidelity envelope that separates transport integrity from evidentiary sufficiency. A claim declares required event types, payload paths, causal relations, observation roles, and integrity state; a multi-valued decision reports supported, weakened, or inconclusive rather than silently interpreting missing evidence as truth. The format uses canonical events, content-addressed blobs, segmented streaming, explicit capture profiles, DSSE-compatible manifests, typed redaction placeholders, and selective-disclosure dependency closure. On a frozen 120-case constructed corpus, the executable mechanism matched all labels; integrity-only and profile-only baselines achieved 0.333 and 0.167 accuracy. Same-team component ablations exposed false support when known integrity invalidity was ignored. Cross-library tests exercised RFC 8785 canonicalization and DSSE verification; official OTLP protobuf, OpenTelemetry Collector, Inspect, OpenInference, AgentRewardBench, and STIX paths were tested with explicit loss accounting. Signed synthetic traces scaled to 50,000 events, but the performance results come from one host. A bounded privacy study confirmed dictionary, equality-correlation, length, bucket, path, and presence leakage under relevant strategies. The construction provides tested tamper evidence and provenance only within its stated threat model; it does not prove truth, completeness, non-repudiation, or recorder honesty.

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1. Problem

An AI behavior conclusion may require:

- a prompt and response;
- an observed tool action and effect;
- a parent relationship showing which request produced an outcome;
- a complete artifact;
- a named capture policy;
- valid package and event-chain integrity;
- a verified trust policy;
- explicit omissions and dropped-event counts.

A trace can be cryptographically intact yet lack one of those requirements. Conversely, a redacted trace may still support a narrow claim that never required the hidden content. A universal "valid trace" Boolean therefore conflates transport integrity with claim support.

2. Proposed fidelity envelope

For each claim, the producer declares typed evidence requirements:

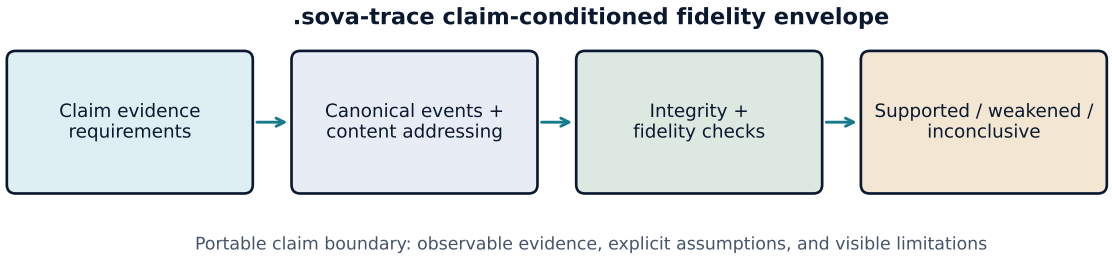


Figure 1: Conceptual architecture for Claim-Conditioned Evidence Fidelity. Boxes show the portable decision boundary; executor or producer mechanics remain outside the claim.

- required event kind and minimum count;
- whether full content is necessary;
- whether redaction is tolerable;
- mandatory or optional status;
- required causal parent kind;
- integrity and signer-trust assumptions.

Each requirement evaluates to:

- `present`;
- `redacted`;
- `omitted`;
- `missing`;
- `invalid`;
- `unverified`.

The claim becomes:

- `supported` when every mandatory requirement is present;
- `weakened` when only explicitly tolerated or optional requirements degrade;
- `inconclusive` when a mandatory requirement is absent, invalid, irreversibly redacted, or outside the supplied trust decision.

The central safety invariant is monotonicity: removing or invalidating required evidence cannot increase claim support.

3. Completeness vocabulary

The design keeps separate:

1. **package continuity** - declared objects match their digests;
2. **event-chain continuity** - covered local events were not changed or reordered;
3. **recorder completeness** - the recorder reports whether it dropped data;
4. **sensor coverage** - the required source was instrumented;
5. **source completeness** - the observed producer emitted the relevant fact;
6. **semantic sufficiency** - the retained evidence can support this claim.

No hash chain proves the absence of unrecorded events. No signature proves that an observation is true.

4. Disclosure graph prototype

A claim identifies seed evidence events. The prototype computes their causal ancestor closure. Every event receives a domain-separated, salted commitment; the ordered commitments are bound into a root. Events in the closure reveal their bytes and nonce, while unrelated events retain only commitments.

The verifier checks:

- sequence and commitment binding;
- revealed event identifiers and bytes;

- the commitment root.

The root must itself be authenticated outside this prototype. Hidden commitments do not prove that hidden plaintext is true, and withholding the nonce prevents the receiver from testing guesses against hidden low-entropy content. This is not claimed as a new commitment scheme, zero-knowledge proof, or production cryptographic selective-disclosure protocol.

5. Prior art and standards boundary

OpenTelemetry (OpenTelemetry Authors 2026b) supplies broad event and trace semantics. W3C PROV (World Wide Web Consortium 2013b, 2013a) formalizes provenance validity and equivalence. in-toto, DSSE, and Sigstore (Torres-Arias et al. 2019; Secure Systems Lab 2023; Sigstore Authors 2026) provide typed attestation and portable verification material. Authenticated logs, Merkle trees, salted redaction proofs, encrypted audit logs, and selective disclosure have extensive paper, standards, and patent prior art. Inspect AI and MLflow already preserve rich AI evaluation and tracing records.

The candidate research delta is not a new trace container or cryptographic primitive. It is machine-checkable propagation from trace transformations and capture loss into the support status of a declared AI-behavior conclusion.

6. Experimental protocol

6.1 Fidelity corpus

The synthetic corpus contains 120 claim/trace combinations with:

- complete evidence;
- redacted required content;
- omitted event families;
- invalid integrity;
- missing causal parents;
- unverified signer policy.

The tested claim requires a tool event linked to a prompt and an oracle event linked to the tool.

6.2 Baselines

- integrity valid alone;
- capture profile name alone;
- full trace sharing without requirement analysis.

6.3 Disclosure mutations

The study mutates:

1. disclosed event content;
2. disclosed nonce;
3. commitment digest;
4. root;
5. sequence.

6.4 Substrate run

One CPU-only reference-host run measured:

- 10,000 compact events;
- 1,000 events with approximately 1 KiB model-response payloads;
- 100 identical artifact additions to exercise deduplication;
- valid-prefix recovery after an incomplete JSON tail.

6.5 Post-result component and independent-verifier ablations

The complete public trace substrate and private claim-fidelity mechanism were then exercised under explicit removal or weakening of:

- the event hash chain after package descriptors and an unsigned manifest digest were recomputed;
- content-addressed blob integrity;
- trusted observer provenance;
- the integrity-to-claim fidelity gate;
- dependency closure during selective disclosure;
- Standard versus Lite capture;
- included-key versus required-key signature policy.

The same signed trace was independently checked by a Python implementation that imports no SOVA module and by a dependency-free Node.js implementation. Both support required-key DSSE/Ed25519 verification. Neither verifies external timestamp/transparency state.

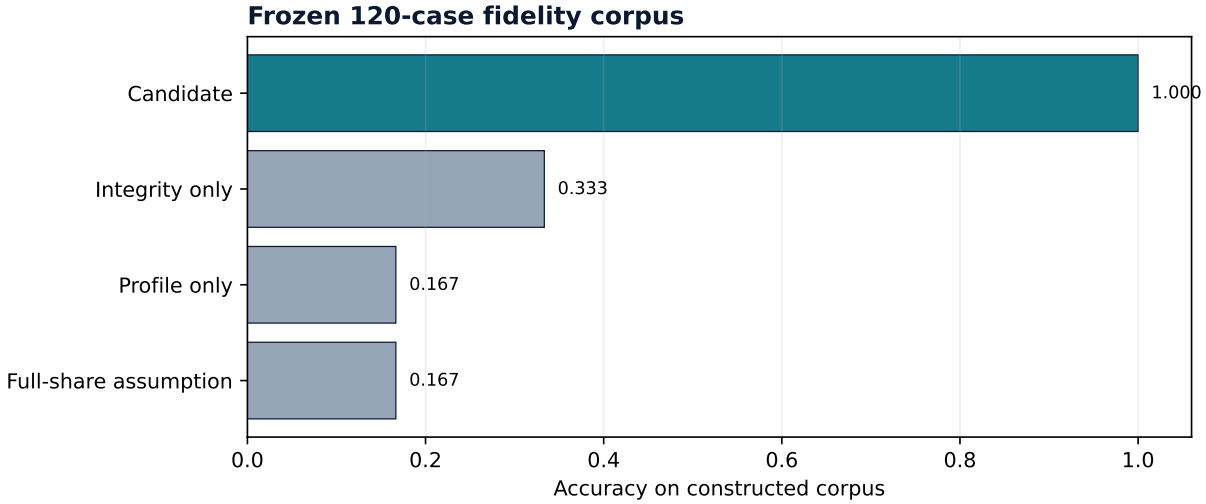


Figure 2: Constructed-corpus baseline comparison. Values are retained experiment results, not field-performance estimates.

7. Results

7.1 Claim support

Method	Accuracy
Claim-conditioned fidelity	1.000
Integrity only	0.333
Capture profile name only	0.167
Full-share assumption	0.167
Remove causal-parent requirements	0.833
Remove full-content requirements	0.833

No constructed case missing mandatory evidence was classified as supported. As with the Topic 04 corpus, the same team wrote the transformations and evaluator; the result shows specification consistency, not external validity. The two component ablations were added after the original run under disclosed protocol amendment 0.3. Each caused false support for the balanced case family whose requirement it removed.

7.2 Disclosure prototype

All five specified mutations were detected. The dependency closure disclosed three of four events (75%) and withheld the event explicitly labeled irrelevant. This tiny case does not establish disclosure minimization, unlinkability, dictionary resistance, or performance at scale.

7.3 Reference-host substrate

Workload	Result
10,000 compact events	1,185.15 events/s
Compact trace package	728,688 bytes
1,000 ~1 KiB responses	1,152.03 events/s
Medium trace package	78,132 bytes
Interrupted prefix recovered	10 events

The package sizes reflect ZIP compression and repetitive synthetic data. Numbers are from one Windows/Python run and are not public performance claims.

7.4 Component-ablation and cross-language results

- A modified event segment was repackaged with fresh object descriptors and a fresh unsigned manifest digest. Package-only checks accepted it; full and independent verifiers rejected the stale event hash.
- Direct blob-byte substitution failed content-descriptor integrity.
- The executor conclusion was `conformant` with a trusted SOVA observer and `inconclusive` without it; provider status alone would have accepted it.
- Known invalid integrity made the claim `inconclusive`. An ablated evaluator that ignored the invalidity returned the false `supported` result.
- Dependency closure included both direct oracle parents; a seed-only disclosure omitted two required parents. The committed disclosure verified and rejected disclosed-content tampering.
- Standard capture supported the claim. Lite dropped the filesystem family and produced `inconclusive`, rather than relying on the profile name.
- Included-key verification remained `included-key-integrity-only`; a pinned key produced `required-key`; a wrong required key failed; carried timestamp material remained unverified.
- Python and Node.js independent verifiers agreed on the signed trace's canonical content digest and rejected the re-manifested chain mutant.

These results make the role of each layer concrete. They remain same-team synthetic checks and are not a new cryptographic construction or privacy proof.

7.5 Pinned external-fixture results

- At Inspect AI commit `1ea01a9e1b3c94db3cdf93ce5583ffebf86e159f`, thirteen official JSON logs contained transcript events, legacy flat events, no-event-stream records, and no-sample logs.
- Adapter 0.2 accepted 5/13 and typed-refused eight. This retained negative result falsified its transcript-only assumption.
- Adapter 0.3 accepted the unchanged 13/13 corpus, converted fourteen samples and 200 source events into 280/280 verified SOVA events, declared every conversion lossy, and produced zero known forbidden-content findings.
- All twenty-two cleaned AgentRewardBench (Lu et al. 2025) WebArena trajectories produced verified traces and capsules using action verb/digest and external categorical labels. Goals, raw reasoning, arguments, screenshots, accessibility trees, URLs, chat messages, and annotator identity were excluded from retained decoded structures.
- The complete 1,408-row annotation table was summarized but not redistributed; the Space metadata did not assert a source-data licence.

7.6 Cross-library integrity and canonicalization

- securesystemslib 1.4.0 independently accepted the SOVA DSSE envelope and rejected payload mutation, signature mutation, a wrong key, key-ID mismatch, and semantically equal non-canonical payload substitution.
- rfc8785.py 0.1.4 produced byte-identical canonical JSON for 65/65 values in SOVA's intentionally shared domain. SOVA rejected a binary float that the full JCS implementation serialized; SOVA does not claim complete RFC 8785 (Rundgren et al. 2020) number support.
- These are cross-library byte/signature checks. They do not establish signer identity, key ownership, freshness, transparency inclusion, truth, completeness, non-repudiation, or unforgeability.

7.7 Official OTLP protobuf boundary

OpenTelemetry protobuf (OpenTelemetry Authors 2026a) 1.44.0 preserved trace ID, span ID, parent, links, name, start time, attributes, event count, and an unknown external attribute. The round trip remained explicitly lossy:

- export omitted SOVA monotonic time, clock record, redactions, attempt, and previous-event hash;
- import could not reconstruct SOVA authorization, safety policy, absent monotonic ordering, capture-time-redaction proof, or the event hash chain.

No collector or hosted backend was used. The result establishes transport compatibility, not semantic equivalence or production backend certification.

7.8 Quantitative privacy-attack study

A retained same-team study added explicit dictionary, linkage, length, structure, compromised-key, nonce, and detector-negative-control cases:

- an unkeyed SHA-256 baseline uniquely recovered a low-entropy synthetic value from a 100-entry dictionary;
- a keyed commitment did not match any naive unkeyed dictionary candidate, but the same dictionary uniquely recovered the value after the HMAC key was treated as compromised;
- the same value produced equal commitments inside one context and different commitments across two contexts, confirming intentional within-context linkability and tested cross-context domain separation;
- unpadded AES-GCM ciphertext length had Pearson correlation `1.000000` with canonical plaintext byte length in the seven-length fixture;
- authenticated power-of-two padding reduced exact length to declared buckets, but bucket, placeholder, path, and surrounding-structure leakage remained;
- omission produced a constant-size marker over the fixture but still exposed presence, class, method, and path records;
- 512 encrypted repetitions produced 512 distinct nonces, while a generic sensitive string without a recognized secret shape remained a deliberate detector false negative.

These results caused an implementation change: encrypted placeholders can now use authenticated ISO/IEC 7816-4-style power-of-two length buckets, and decryption rejects padding metadata that differs from authenticated associated data. The experiment does not prove HMAC, AES-GCM, anonymity, or malicious- recorder resistance. It strengthens the policy decision that public export defaults to omission.

These results improve schema, privacy, and interoperability falsification. They do not demonstrate complete Inspect fidelity, general cross-format equivalence, expert-judge calibration, or privacy against linkage and inference from action digests.

7.9 Incident interchange, official Collector, and scale envelope

The incident path projected one safe eight-event SOVA trace into an OASIS STIX (OASIS Open 2021) 2.1 bundle containing one each of `identity`, `incident`, `artifact`, `observed-data`, `relationship`, and `note`. The official `stix2==3.0.2` parser, pinned at commit `679bfe0befae5c39143cedcdc3cacf97ac31c890`, accepted the bundle. Import created a seven-event trace that passed offline verification and retained custom references to the source trace digest and chain root. Fidelity remained explicitly lossy: STIX did not preserve the raw event stream, full causal graph, authorization, redaction history, replay procedure, environment requirements, or DSSE trust policy.

For the backend path, the official `otelcol-contrib` v0.157.0 Windows archive matched the published checksum `7b3938e1522ff04261a694a58e7111c5f7cdd19be617c3a77118cffad7abb815`. The extracted binary had digest `sha256:76146071912cc973553058795b7bb1b0c839991757257dabfce2d99fa89e26be`. A local-only Collector configuration received eight OTLP/HTTP protobuf spans on `127.0.0.1` and wrote eight records through the official file exporter. Trace IDs, span IDs, and required SOVA projection attributes were preserved. This establishes one real Collector path, not hosted-vendor certification, production reliability, privacy, or semantic equivalence.

The scale envelope produced signed traces with 1,000, 10,000, and 50,000 events and packages of 126,203, 1,238,352, and 6,183,508 bytes in the retained run. Every case passed required-key verification, indexed

queries, 100-event payload-minimized disclosure, and dependency-free Python/Node content-digest agreement. The 50,000-event case used fifty segments, captured at 325.6 events/s under instrumentation, and reached 54,973,724 bytes of traced peak memory. Those timings are diagnostic one-host samples, not product performance claims. The largest case is 56.88 times the 879 events retained across the two external corpora, but it is synthetic and does not substitute for independently operated external-corpus scale.

8. Security analysis

The prototype and public trace foundation are tested against malformed archives, traversal, duplicate names, excessive compression, event insertion, deletion, reordering, substitution, duplicate identifiers, forward causal parents, invalid signatures, signer-policy mismatch, malformed redaction placeholders, secret-shaped fields, and truncated tails.

Residual threats include:

- a dishonest or compromised recorder;
- an uninstrumented source;
- colluding signer and recorder;
- semantic omissions not declared by the producer;
- correlation, bucket-size, path, and surrounding-structure leakage;
- low-entropy guessing if salts or keys are exposed;
- malicious but schema-valid extension payloads;
- unverifiable external timestamps or transparency data;
- claims whose evidence requirements were authored incorrectly.

9. Threats to validity

Construct validity

The envelope evaluates author-declared evidence requirements. An incorrect or incomplete claim specification can produce a mechanically correct but scientifically misleading conclusion. Event presence is not observation truth, and a trace may omit an uninstrumented effect without detectable discontinuity.

Internal validity

The SOVA team authored the mechanism, constructed corpus, most converters, and interpretation. Perfect synthetic labels primarily show executable-specification consistency. Independent libraries and external corpora reduce implementation coupling but do not remove same-team selection or analysis bias.

External validity

The current external corpus is bounded to Inspect JSON and `.eval`, privacy- minimized AgentRewardBench trajectories, canonical/signature libraries, one OASIS STIX incident projection, one local official Collector/file-exporter path, and an OTLP protobuf round trip. It excludes hosted production backends, diverse vendor traces, real incident corpora, malicious recorders, and independently operated environments.

Statistical validity

The 120 cases are deterministic and balanced by construction. Reference-host throughput and size measurements, including the 50,000-event synthetic envelope, have no multi-machine variance estimate. The external study must report denominators, per-stratum uncertainty, and failures rather than aggregate success percentages alone.

Security and privacy validity

The same-team attack matrix is not a cryptographic audit. Signatures and hash chains provide bounded tamper evidence, not truth, freshness, completeness, non-repudiation, or signer honesty. Authenticated padding still leaks buckets, paths, presence, and surrounding structure; omission remains the safest public default for secrets.

10. Future independent validation

A confirmatory evaluation must use independently produced traces and claim contracts, a cryptographic and privacy review, malicious-recorder cases, differential parsing, hosted or independently operated telemetry backends, and established authenticated-log and selective-disclosure baselines. The retained reviewer packet freezes attack cases, source-format families, adverse findings, digests, and scoring rules. It verifies handoff integrity but does not establish reviewer independence.

11. Conclusion

A trace can be byte-intact and still be insufficient for a behavioral claim. The claim-conditioned fidelity envelope makes this distinction executable and propagates known gaps instead of converting them into unsupported certainty. The current evidence supports a bounded engineering and research result; independent security, privacy, and external-corpus studies remain necessary before broader claims.

Declarations

Ethics and dual-use statement

The reported machine experiments use synthetic or public research fixtures and safe local actions. They do not target a live third party, use client data, store real credentials, or claim access to private model thoughts or hidden chain-of-thought. The work can support security testing and therefore has dual-use potential. Unsafe reproduction remains subject to explicit authorization, containment, and human review; operational exploit details and novelty-sensitive implementations are not part of this private review package.

Data and artifact availability

The generic SOVA implementation, public specifications, safe examples, and tests are available in the SOVA-OSS repository. Private experiment results, preregistrations, source digests, build provenance, and review packets are retained by TRUSCOR Private Limited. Third-party source artifacts whose redistribution rights are unclear are not redistributed. A release artifact must be assembled only after rights, privacy, security, and IP review.

Author contributions and AI-assisted work

Gautam R. Patil conceived the SOVA project, specified the research direction and safety constraints, and directed this work. Software implementation, experiment execution, analysis support, literature discovery, and drafting used OpenAI Codex under his direction. Gautam R. Patil is the sole proposed human author and must verify the final text, citations, figures, data, and venue disclosure before submission. No AI system is listed as an author or inventor.

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This work was conducted for TRUSCOR Private Limited. No external funding was used for the reported machine experiments. The author's affiliation with TRUSCOR Private Limited is a relevant non-financial and potential commercial interest. The exact legal entity name and any additional interests must be confirmed before submission.

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